CIVE70108 Assignment – Supply Chain Game

This assignment is a web-based discrete-event simulation of the supply chain that consists of a network of factories and warehouses in a fictitious continent. As a team of logisticians, you must propose an inventory strategy that maximises profit over the simulation period.

The game is set up such that each drum of cargo sells at $1,450 and 10% interest per year on your capital. The expenses are made up of:

- Production costs – which equate to $2,000 fixed and $900 variable (per item).
- Shipping costs – which depend on the type of transportation used. $150 per drum for mail and $15,000 per truck which holds up to 200 drums.
- Finished goods holding costs – which equate to $90 per drum per year.
- Customer fulfilment costs – equating to $150 per drum.
- Factory capacity costs – $50,000 per drum per day.

Week 3 introduces a larger problem instance, which includes additional costs to consider:

- $100,000 for every new warehouse built.
- $500,000 for every new factory built.

Please register your team **before January 23rd!**

Stage 1 – Trial Run

The simulation starts on the January 26th and focuses on a One Region scenario. This serves as a **trial run** that will allow you to become more familiar with the problem and the interface. A description on this scenario can be seen by in the following address: op.responsive.net/sc/escribano/Assign1. This is not a link, please copy and paste it on your browser of preference.

Stage 2 – Assessed Run 1 (One Region)

Starting on the 3rd of February, the graded One Region scenario will run. A description on this scenario can be seen by in the following address: op.responsive.net/sc/escribano/Assign1. This is not a link, please copy and paste it on your browser of preference.

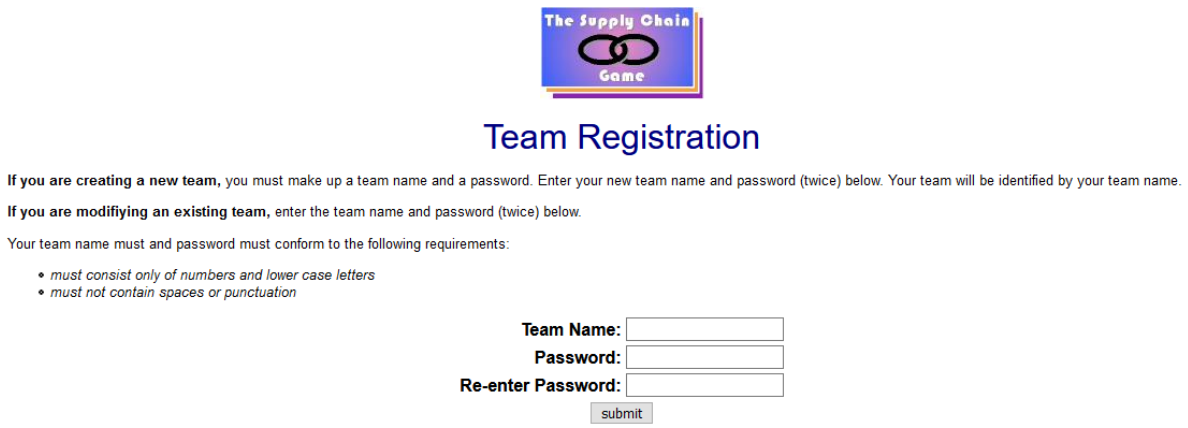
Stage 3 – Assessed Run 2 (Network)

Starts on the 17th of February, this problem instance includes 5 different regions and more decisions to make, including building new factories and/or warehouses. A description on this scenario can be seen by in the following address: op.responsive.net/sc/escribano/Assign2. This is not a link, please copy and paste it on your browser of preference.

NOTE: the values on the description pages above will be different to the ones in this exercise. Refer to the values in this document! This applies to all runs.

REGISTER

Each group will have to register a team through the following URL: op.responsive.net/sc/escribano/start.html. Use the code 'transport' to access the team registration page, which should look like this:



Register your team by introducing the team's name and password.

If you want to modify your team, you can enter the existing team's name and password.

You must add at least one Team Member name, otherwise the team will not be created. There is no need to add all team members.

In the team registration page, there will be an option to include a section number that says: *"If there is more than one section of your course, enter section number here:"*. **Please make sure to leave this as "0".**